Poisson Regression

STA 310: Homework 4 - Amaris

Instructions:

- Write all narrative using full sentences. Write all interpretations and conclusions in the context of the data.
- Be sure all analysis code is displayed in the rendered pdf.
- If you are fitting a model, display the model output in a neatly formatted table. (The tidy and kable functions can help!)
- If you are creating a plot, use clear and informative labels and titles.
- Make sure to upload to both Gradescope and Canvas in a reproducible format per the instructions of prior homework assignments.

These exercises are derived from BMLR, Chapter 4.

1.

Answer parts a - d in the context of the following study:

A state wildlife biologist collected data from 250 park visitors as they left at the end of their stay. Each was asked to report the number of fish they caught during their one-week stay. On average, visitors caught 21.5 fish per week.

a. **Define the response.**

   The response variable is the number of fish caught by a park visitor during their one-week stay.

b. **What are the possible values for the response?**

   The response takes non-negative integer values $(0,1,2,3,\ldots)$ since fish counts cannot be negative or fractional.

c. **What does λ represent?**

   In a Poisson model, λ represents the expected (mean) number of fish caught per visitor per week. Here, based on the sample, $\lambda = 21.5$.

d. **Would a zero-inflated model be considered here? If so, what would be a "true zero"?**

   Yes, a zero-inflated model could be considered if there are more zeros than expected under a standard Poisson model. A true zero would correspond to visitors who did not fish at all during their stay (structural zeros), rather than visitors who fished but happened to catch zero fish by chance.

2.

@brockmann1996 **carried out a study of nesting female horseshoe crabs. Female horseshoe crabs often have male crabs attached to a female's nest known as satellites. One objective of the study was to determine which characteristics of the female were associated with the number of satellites. Of particular interest is the relationship between the width of the female carapace and satellites.**

The data can be found in crab.csv in the data folder. It includes the following variables:

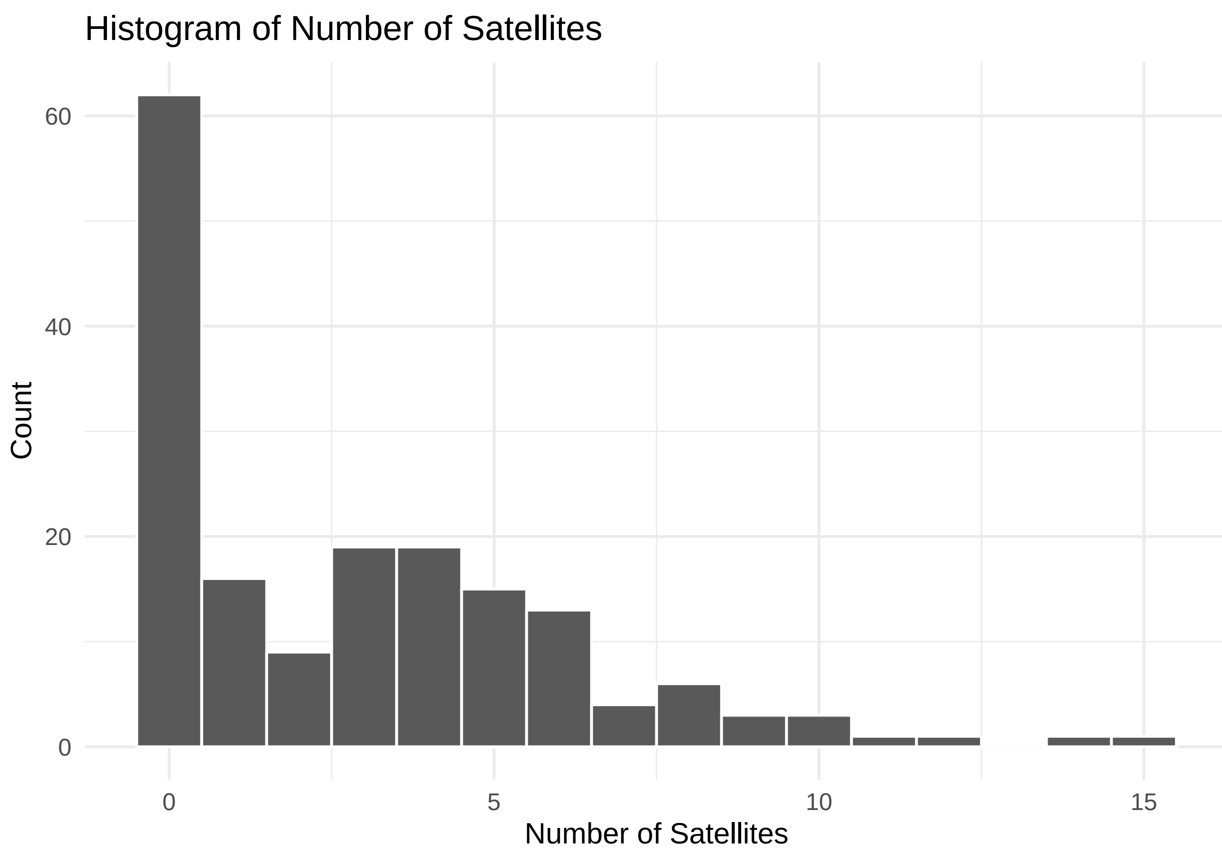
- **Satellite = number of satellites**
- **Width = carapace width (cm)**
- **Weight = weight (kg)**
- **Spine = spine condition (1 = both good, 2 = one worn or broken, 3 = both worn or broken)**
- **Color = color (1 = light medium, 2 = medium, 3 = dark medium, 4 = dark)**

Make sure to convert Spine and Color to the appropriate data types in R before doing the analysis.

```
crab <- read.csv("crab.csv")
library(ggplot2)
library(broom)
library(dplyr)
library(knitr)
```

a. **Create a histogram of Satellite. Is there preliminary evidence the number of satellites could be modeled as a Poisson response? Briefly explain.**

```
crab <- crab |>
  mutate(
    Spine = factor(Spine, levels = c(1, 2, 3),
                   labels = c("both_good", "one_worn_broken", "both_worn_broken")),

    Color = factor(Color, levels = c(1, 2, 3, 4),
                   labels = c("light_medium", "medium", "dark_medium", "dark"))
  )


ggplot(crab, aes(x = Satellite)) +
  geom_histogram(binwidth = 1, color = "white") +
  labs(
    title = "Histogram of Number of Satellites",
    x = "Number of Satellites",
    y = "Count"
  ) +
  theme_minimal()
```



Yes — there is preliminary evidence that the number of satellites could be modeled as a Poisson response.

The histogram shows that Satellite is a count variable taking non-negative integer values, with a right-skewed distribution and many small counts, which is consistent with a Poisson-type process. However, there is also a large spike at zero and a long right tail, suggesting possible overdispersion or excess zeros, so while a Poisson model is a reasonable starting point, it may not fully capture the variability in the data.

b. **Fit a Poisson regression model including Width, Weight, and Spine as predictors. Display the model with the 95% confidence interval for each coefficient.**

```
# Fit Poisson regression
crab_pois <- glm(
  Satellite ~ Width + Weight + Spine,
  family = poisson(link = "log"),
  data = crab
)

# Display model summary
crab_pois |>
  tidy(conf.int = TRUE, conf.level = 0.95)|>
  mutate(across(where(is.numeric), round, 3)) |>
  kable( caption = "Poisson Regression of Number of Satellites on Female Characteristics",
         col.names = c("Term",
                       "Estimate",
                       "Std. Error",
                       "z value",
                       "p-value",
```

```
                       "Lower 95% CI", "Upper 95% CI"
                       )
          )
```

Table 1: Poisson Regression of Number of Satellites on Female Characteristics

Term	Estimate	Std. Error	z value	p-value	Lower 95% CI	Upper 95% CI
(Intercept)	-1.062	0.928	-1.144	0.253	-2.875	0.763
Width	0.039	0.048	0.816	0.415	-0.055	0.132
Weight	0.000	0.000	2.771	0.006	0.000	0.001
Spineone_worn_broken	-0.214	0.211	-1.017	0.309	-0.644	0.185
Spineboth_worn_broken	-0.049	0.108	-0.458	0.647	-0.257	0.165

c. **Describe the effect of Spine in terms of the mean number of satellites.**

Because this is a Poisson regression with a log link, the coefficient for Spine represents a multiplicative effect on the mean number of satellites. In this Poisson regression model, spine condition is treated as a categorical variable, with "both good" serving as the reference group. The coefficient for one worn/broken spine is -0.214, which implies that the expected mean number of satellites is multiplied by 0.81. Thus, holding width and weight constant, female crabs with one worn/broken spine are expected to have about 19% fewer satellites on average compared to female crabs with both spines in good condition.

Similarly, the coefficient for both worn/broken spines is -0.049, corresponding to a multiplicative change of 0.95. This suggests that, controlling for width and weight, female crabs with both spines worn or broken are expected to have about 5% fewer satellites on average compared to female crabs with both spines in good condition. Overall, while the estimated effects are negative, the p-values indicate that spine condition is not a statistically significant predictor of the mean number of satellites in this model.

3. **Use the scenario from the previous exercise to answer questions (a) - (d).**

a. **We would like to fit a quasi-Poisson regression model for this data. Briefly explain why we may want to consider fitting a quasi-Poisson regression model for this data.**

   A quasi-Poisson regression is appropriate when there is over-dispersion in count data—that is, when the variance of the response exceeds its mean, violating the Poisson assumption $\mathrm{Var}(Y) = \mathbb{E}(Y)$. From the histogram and earlier Poisson fit, the satellite counts show a large number of zeros and a long right tail, suggesting extra variability that a standard Poisson model may not adequately capture. The quasi-Poisson model relaxes the variance assumption while keeping the same mean structure.

b. **Fit a quasi-Poisson regression model that corresponds with the model chosen in the previous exercise. Display the model.**

```
# Fit quasi-Poisson regression
crab_qpois <- glm(
  Satellite ~ Width + Weight + Spine,
  family = quasipoisson(link = "log"),
  data = crab
)

# Display model results with 95% CIs
```

```r
crab_qpois|>
  tidy(conf.int = TRUE, conf.level = 0.95)|>
  mutate(across(where(is.numeric), round, 3)) |>
  kable(
    caption = "Quasi-Poisson Regression of Number of Satellites on Female Characteristics",
    col.names = c("Term", "Estimate", "Std. Error", "t value", "p-value",
                  "Lower 95% CI", "Upper 95% CI")
  )
```

Table 2: Quasi-Poisson Regression of Number of Satellites on Female Characteristics

Term	Estimate	Std. Error	t value	p-value	Lower 95% CI	Upper 95% CI
(Intercept)	-1.062	1.652	-0.643	0.521	-4.281	2.195
Width	0.039	0.085	0.458	0.647	-0.130	0.203
Weight	0.000	0.000	1.556	0.122	0.000	0.001
Spineone_worn_broken	-0.214	0.375	-0.571	0.568	-1.006	0.480
Spineboth_worn_broken	-0.049	0.192	-0.257	0.797	-0.416	0.337

c. **What is the estimated dispersion parameter? Show how this value is calculated.**

```r
# Estimated dispersion parameter
summary(crab_qpois)$dispersion
```

```
## [1] 3.169448
```

It is calculated through: $\hat{\phi} = \dfrac{\sum_{i=1}^{n} r_{P,i}^2}{n-p}$

The dispersion parameter is estimated by taking the sum of squared Pearson residuals and dividing by the residual degrees of freedom (n-p). This quantity measures how much the observed variability in the response exceeds the variability assumed under a standard Poisson model, with values greater than 1 indicating overdispersion. To verify:

```r
sum(residuals(crab_qpois, type = "pearson")^2) / df.residual(crab_qpois)
```

```
## [1] 3.169448
```

d. **How do the estimated coefficients change compared to the model chosen in the previous exercise? How do the standard errors change?**

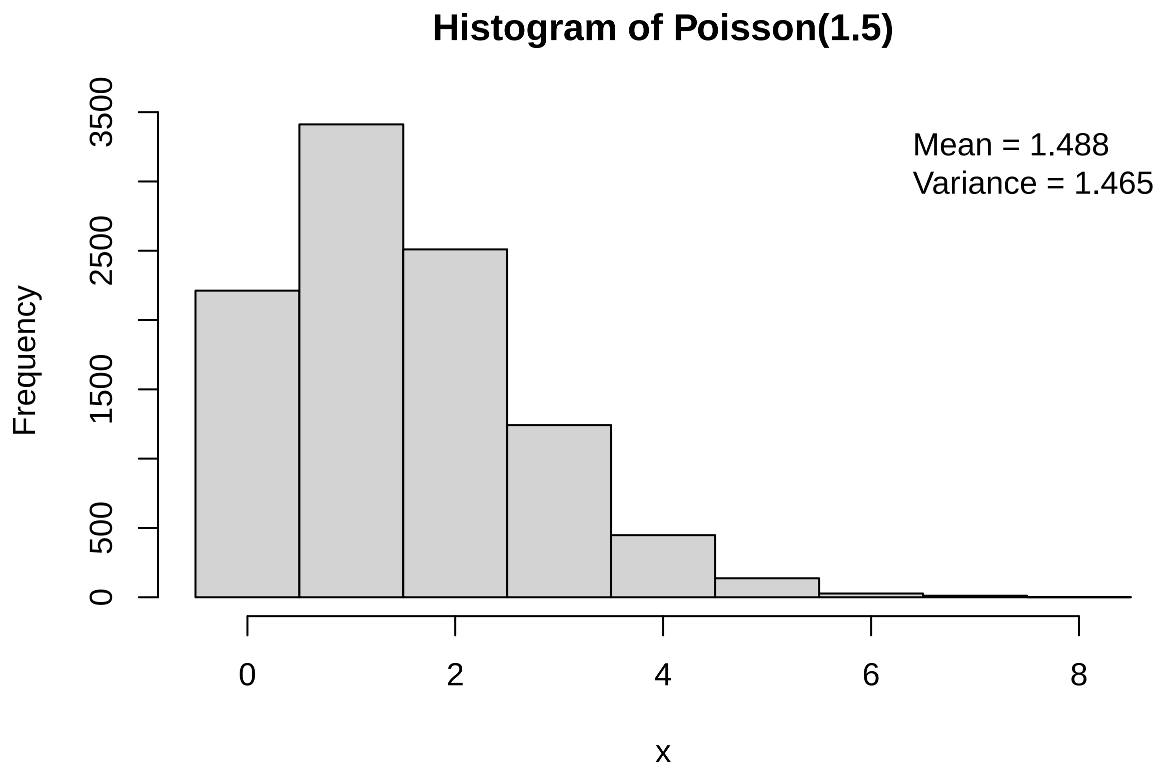
The estimated coefficients are identical in the Poisson and quasi-Poisson models, so the estimated effects of width, weight, and spine on the expected number of satellites do not change. This is because both models use the same functional form to model the mean response. However, the standard errors are larger in the quasi-Poisson model because they are explicitly scaled to account for over-dispersion in the data. Specifically, the quasi-Poisson model inflates the Poisson standard errors by a factor related to the square root of the estimated dispersion parameter, reflecting the fact that the variability in the data is greater than what the Poisson model assumes. This scaling increases uncertainty in the estimates, resulting in wider confidence intervals and larger p-values, and therefore leads to more conservative statistical inference.

4. **The goal of this exercise is to use simulation to understand the equivalency between a gamma-Poisson mixture and a negative binomial distribution.**

Remember to set a seed so your simulations are reproducible!

a. **Use the R function rpois() to generate 10,000 x_i from a regular Poisson distribution, $X \sim \text{Poisson}(\lambda = 1.5)$. Plot a histogram of this distribution and note its mean and variance. Next, let $Y \sim \text{Gamma}(r = 3, \lambda = 2)$ and use rgamma() to generate 10,000 random y_i from this distribution.**

```r
set.seed(123)
n <- 10000

# Generate 10,000 Poisson observations
x <- rpois(n, lambda = 1.5)

# Compute mean and variance
x_mean <- mean(x)
x_var  <- var(x)

# Histogram
hist(
  x,
  breaks = seq(-0.5, max(x) + 0.5, by = 1),
  main = "Histogram of Poisson(1.5)",
  xlab = "x",
  col = "lightgray"
)

# Add mean and variance to the plot
legend(
  "topright",
  legend = c(
    paste("Mean =", round(x_mean, 3)),
    paste("Variance =", round(x_var, 3))
  ),
  bty = "n"
)
```

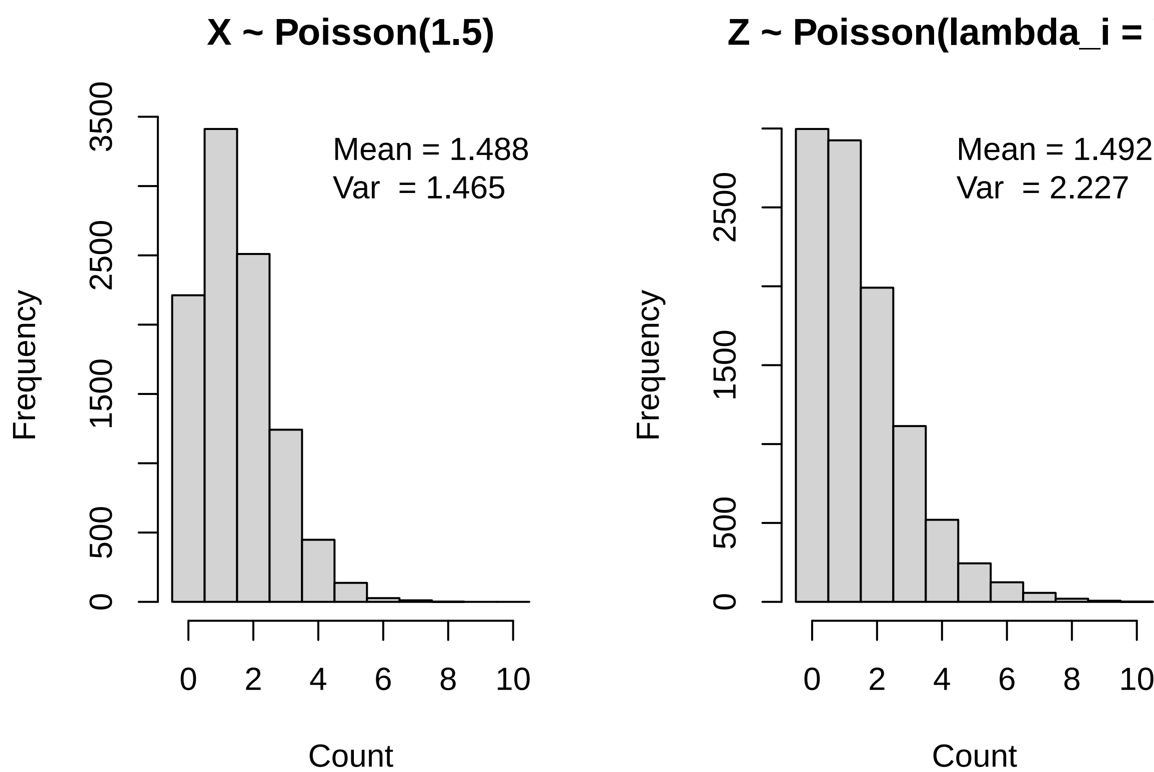



```r
# Generate 10,000 Gamma observations
y <- rgamma(n, shape = 3, rate = 2)
```

Now, consider 10,000 different Poisson distributions where $\lambda_i = y_i$. Randomly generate one z_i from each Poisson distribution. Plot a histogram of these z_i and compare it to your original histogram of X (where $X \sim \text{Poisson}(1.5)$). How do the means and variances compare?

```r
# Mixture counts: Z_i | Y_i ~ Poisson(lambda_i = Y_i)
z <- rpois(n, lambda = y)

# Means and variances
x_mean <- mean(x); x_var <- var(x)
z_mean <- mean(z); z_var <- var(z)

# Use common breaks so histograms are comparable
brks <- seq(-0.5, max(c(x, z)) + 0.5, by = 1)

par(mfrow = c(1, 2))

hist(x, breaks = brks, col = "lightgray",
     main = "X ~ Poisson(1.5)", xlab = "Count")
legend("topright",
       legend = c(paste("Mean =", round(x_mean, 3)),
                  paste("Var  =", round(x_var, 3))),
       bty = "n")
```

```r
hist(z, breaks = brks, col = "lightgray",
     main = "Z ~ Poisson(lambda_i = Y_i)", xlab = "Count")
legend("topright",
       legend = c(paste("Mean =", round(z_mean, 3)),
                  paste("Var  =", round(z_var, 3))),
       bty = "n")
```




```r
par(mfrow = c(1, 1))
```

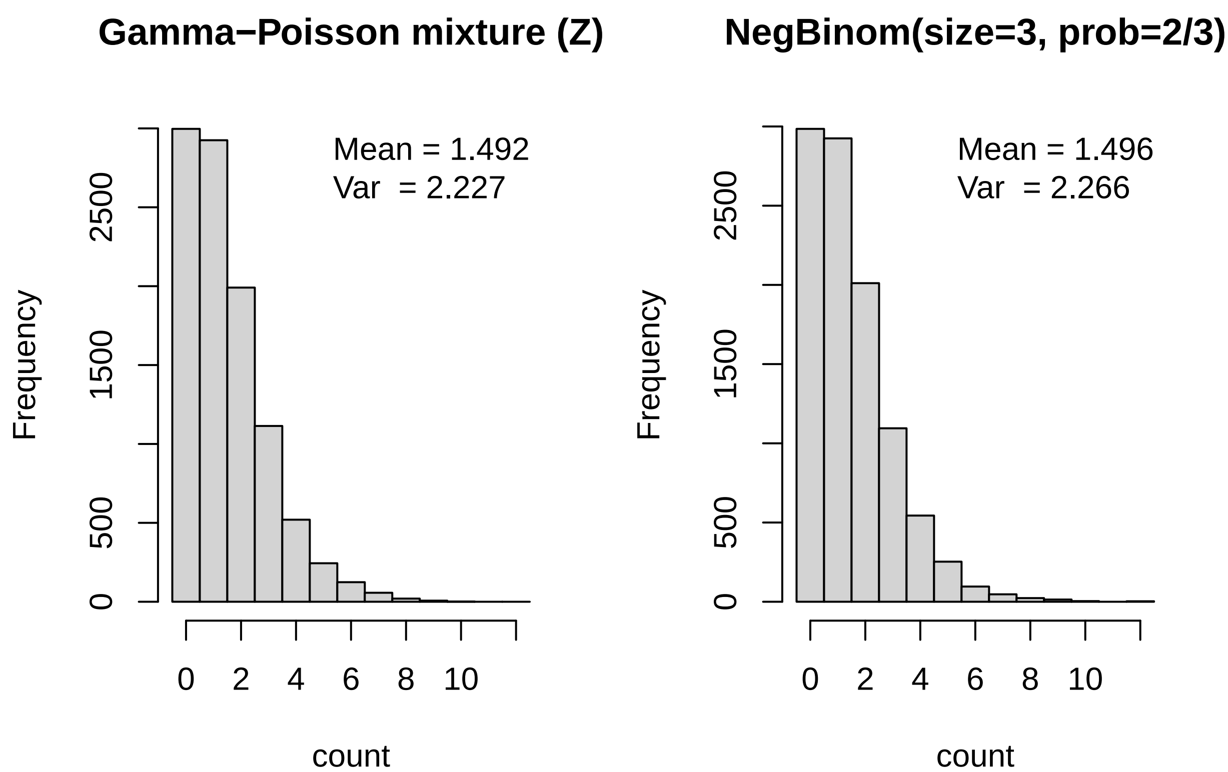
The means of the two distributions are very similar, both being close to 1.5. However, the variance of the gamma–Poisson mixture is substantially larger than the variance of the standard Poisson distribution. This occurs because the mixture includes additional variability from the random Poisson rates, whereas the standard Poisson distribution assumes a fixed rate and therefore has variance approximately equal to its mean.

b. A negative binomial distribution can actually be expressed as a gamma-Poisson mixture. In Part a, you looked at a gamma-Poisson mixture $Z \sim \text{Poisson}(\lambda)$ where $\lambda \sim \text{Gamma}(r = 3, \lambda' = 2)$.

Find the parameters of a negative binomial distribution $X \sim \text{NegBinom}(r, p)$ such that X is equivalent to Z. As a hint, the means of both distributions must be the same, so $\frac{r(1-p)}{p} = \frac{3}{2}$.

Show through histograms and summary statistics that your negative binomial distribution is equivalent to the gamma-Poisson mixture. You can use rnbinom() in R.

```r
# Given parameters from Gamma distribution
r <- 3
mean_lambda <- 3 / 2

# Solve for p
p <- r / (r + mean_lambda)

# Equivalent Negative Binomial: X ~ NegBinom(r=3, p=2/3)
x_nb <- rnbinom(n, size = 3, prob = p)

# Summary statistics
nb_mean <- mean(x_nb); nb_var <- var(x_nb)

# Common breaks for comparable histograms
brks <- seq(-0.5, max(c(z, x_nb)) + 0.5, by = 1)

par(mfrow = c(1, 2))

hist(z, breaks = brks, col = "lightgray",
     main = "Gamma-Poisson mixture (Z)", xlab = "count")
legend("topright",
       legend = c(paste("Mean =", round(z_mean, 3)),
                  paste("Var  =", round(z_var, 3))),
       bty = "n")

hist(x_nb, breaks = brks, col = "lightgray",
     main = "NegBinom(size=3, prob=2/3)", xlab = "count")
legend("topright",
       legend = c(paste("Mean =", round(nb_mean, 3)),
                  paste("Var  =", round(nb_var, 3))),
       bty = "n")
```




```
par(mfrow = c(1, 1))
```

c. **Make an argument that if you want a** $\mathrm{NegBinom}(r, p)$ **random variable, you can instead sample from a Poisson distribution, where the** λ **values are themselves sampled from a gamma distribution with parameters** r **and** $\lambda' = \frac{p}{1-p}$**. You may show equivalency via the simulations or mathematically, however, make sure your arguments are precise and clear.**

Suppose $\Lambda \sim \mathrm{Gamma}(r, \lambda')$ with shape parameter r and rate parameter λ', and conditional on Λ, let $X \mid \Lambda \sim \mathrm{Poisson}(\Lambda)$. We derive the marginal distribution of X.

By the law of total probability, $\mathbb{P}(X = x) = \int_0^\infty \mathbb{P}(X = x \mid \Lambda = \lambda)\, f_\Lambda(\lambda)\, d\lambda$.

The conditional Poisson probability mass function is $\mathbb{P}(X = x \mid \Lambda = \lambda) = \dfrac{e^{-\lambda}\lambda^x}{x!}$, and the Gamma density (shape r, rate λ') is $f_\Lambda(\lambda) = \dfrac{(\lambda')^r}{\Gamma(r)}\lambda^{r-1}e^{-\lambda'\lambda}$ for $\lambda > 0$.

Substituting these expressions yields $\mathbb{P}(X = x) = \int_0^\infty \dfrac{e^{-\lambda}\lambda^x}{x!} \cdot \dfrac{(\lambda')^r}{\Gamma(r)}\lambda^{r-1}e^{-\lambda'\lambda}\, d\lambda$.

Combining like terms gives $\mathbb{P}(X = x) = \dfrac{(\lambda')^r}{x!\,\Gamma(r)} \int_0^\infty \lambda^{x+r-1}e^{-(1+\lambda')\lambda}\, d\lambda$.

Using the Gamma integral identity $\int_0^\infty \lambda^{k-1}e^{-a\lambda}\, d\lambda = \dfrac{\Gamma(k)}{a^k}$ for $k > 0$ and $a > 0$, with $k = x + r$ and $a = 1 + \lambda'$, we obtain $\mathbb{P}(X = x) = \dfrac{\Gamma(x + r)}{x!\,\Gamma(r)} \cdot \dfrac{(\lambda')^r}{(1 + \lambda')^{x+r}}$.

Now let $\lambda' = \dfrac{p}{1-p}$. Then $1+\lambda' = 1+\dfrac{p}{1-p} = \dfrac{1}{1-p}$, $\dfrac{\lambda'}{1+\lambda'} = \dfrac{p/(1-p)}{1/(1-p)} = p$, and $\dfrac{1}{1+\lambda'} = 1-p$.

Substituting these into the expression above gives $\mathbb{P}(X=x) = \dfrac{\Gamma(x+r)}{x!\,\Gamma(r)}\,p^r(1-p)^x$.

For integer r, we note that $\dfrac{\Gamma(x+r)}{x!\,\Gamma(r)} = \binom{x+r-1}{x}$.

Therefore, $\mathbb{P}(X=x) = \binom{x+r-1}{x}(1-p)^x p^r$ for $x = 0, 1, 2, \ldots$.

This is exactly the probability mass function of a NegBinom(r,p) random variable. Hence, sampling $\Lambda \sim \text{Gamma}\left(r, \lambda' = \dfrac{p}{1-p}\right)$ and then sampling $X \mid \Lambda \sim \text{Poisson}(\Lambda)$ produces a negative binomial distribution.

With a large n, the histograms and summary statistics will be extremely close, which empirically supports the proven equivalence:

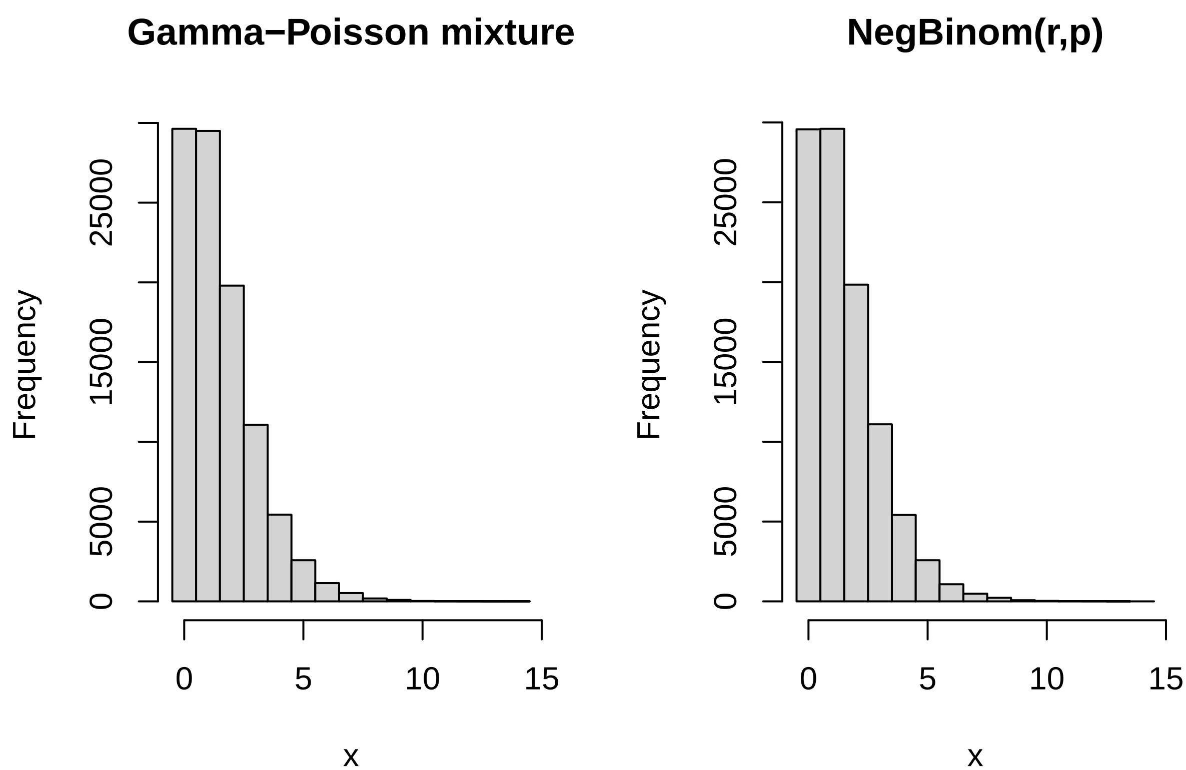
```r
set.seed(456)
n <- 100000

r <- 3
p <- 2/3
lambda_prime <- p/(1 - p)   # rate for Gamma

# Mixture sample
lambda <- rgamma(n, shape = r, rate = lambda_prime)
x_mix <- rpois(n, lambda = lambda)

# Direct NegBinom sample (R parameterization: size=r, prob=p)
x_nb <- rnbinom(n, size = r, prob = p)

# Compare mean/variance
c(mean(x_mix), var(x_mix))
```

```
## [1] 1.50169 2.24602
```

```r
c(mean(x_nb),  var(x_nb))
```

```
## [1] 1.498880 2.233381
```

```r
# Compare histograms
brks <- seq(-0.5, max(c(x_mix, x_nb)) + 0.5, by = 1)
par(mfrow = c(1,2))
hist(x_mix, breaks = brks, col="lightgray", main="Gamma-Poisson mixture", xlab="x")
hist(x_nb,  breaks = brks, col="lightgray", main="NegBinom(r,p)", xlab="x")
```




```r
par(mfrow = c(1,1))
```

5. **In a 2018 study, Chapp et al. (2018) scraped every issue statement from webpages of candidates for the U.S. House of Representatives, counting the number of issues candidates commented on and scoring the level of ambiguity of each statement. We will focus on the issue counts, and determining which attributes (of both the district as a whole and the candidates themselves) are associated with candidate silence (commenting on 0 issues) and a willingness to comment on a greater number of issues. The data set is in ambiguity.csv in the data folder . This analysis will focus on the following variables:**

- **name : candidate name**
- **distID : unique identification number for Congressional district**
- **ideology : candidate left-right orientation**
- **democrat : 1 if Democrat, 0 if Republican**
- **totalIssuePages : number of issues candidates commented on (response)**

See @roback2021beyond for the full list of variables.

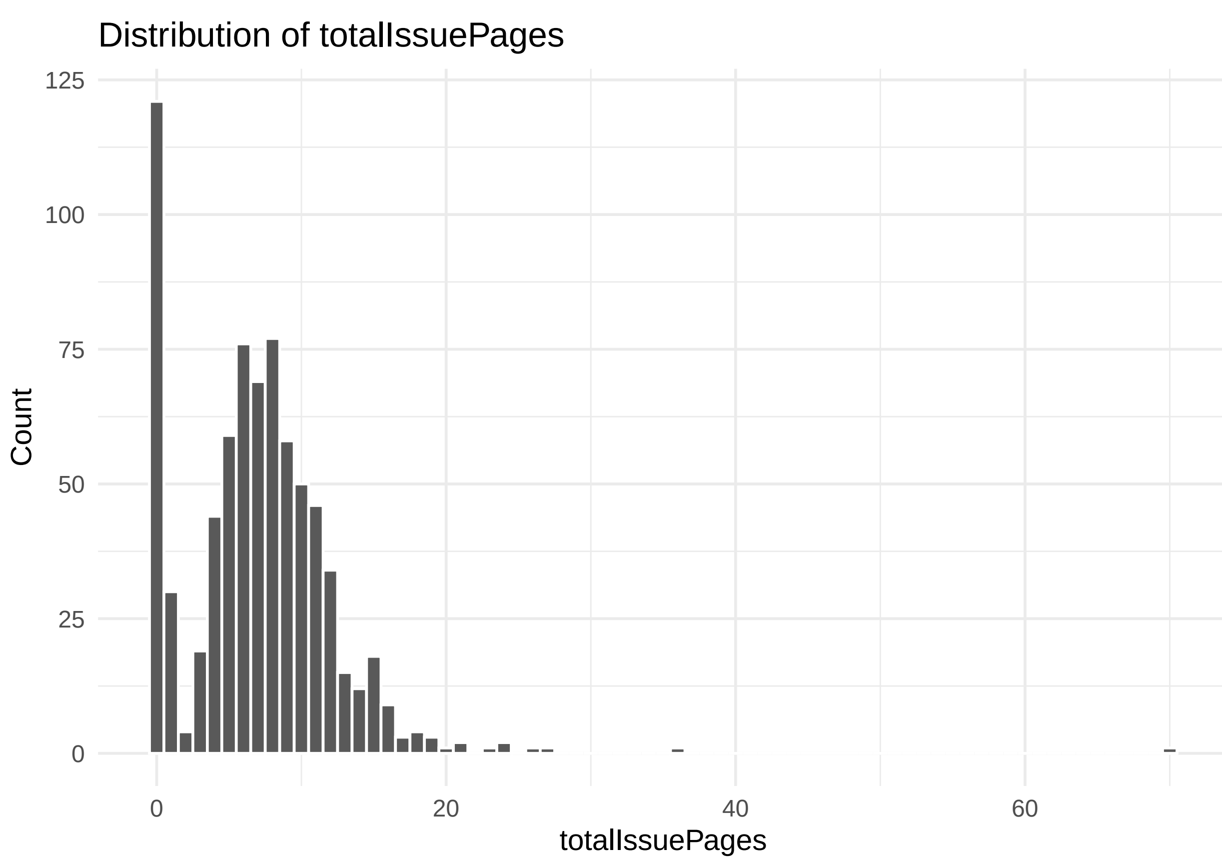
We will use a hurdle model to analyze the data. A hurdle model is similar to a zero-inflated Poisson model, but instead of assuming that "zeros" are comprised of two distinct groups—those who would always be 0 and those who happen to be 0 on this occasion—the hurdle model assumes that "zeros" are a single entity. Therefore, in a hurdle model, cases are classified as

either "zeros" or "non-zeros", where "non-zeros" *hurdle* the 0 threshold—they must always have counts of 1 or above.

We will use the `pscl` package and the `hurdle` function in it to analyze a hurdle model. Note that coefficients in the "zero hurdle model" section of the output relate predictors to the log-odds of being a *non-zero* (i.e., having at least one issue statement), which is opposite of the ZIP model.

a. **Visualize the distribution of the response variable totalIssuePages. Why might we consider using a hurdle model compared to a Poisson model? Why is a zero-inflated Poisson model not appropriate in this scenario?**

```r
library(pscl)
amb <- read.csv("ambiguity.csv")

# Histogram (count-style bins)
ggplot(amb, aes(x = totalIssuePages)) +
  geom_histogram(binwidth = 1, boundary = -0.5, color = "white") +
  labs(x = "totalIssuePages", y = "Count",
       title = "Distribution of totalIssuePages") +
  theme_minimal()
```

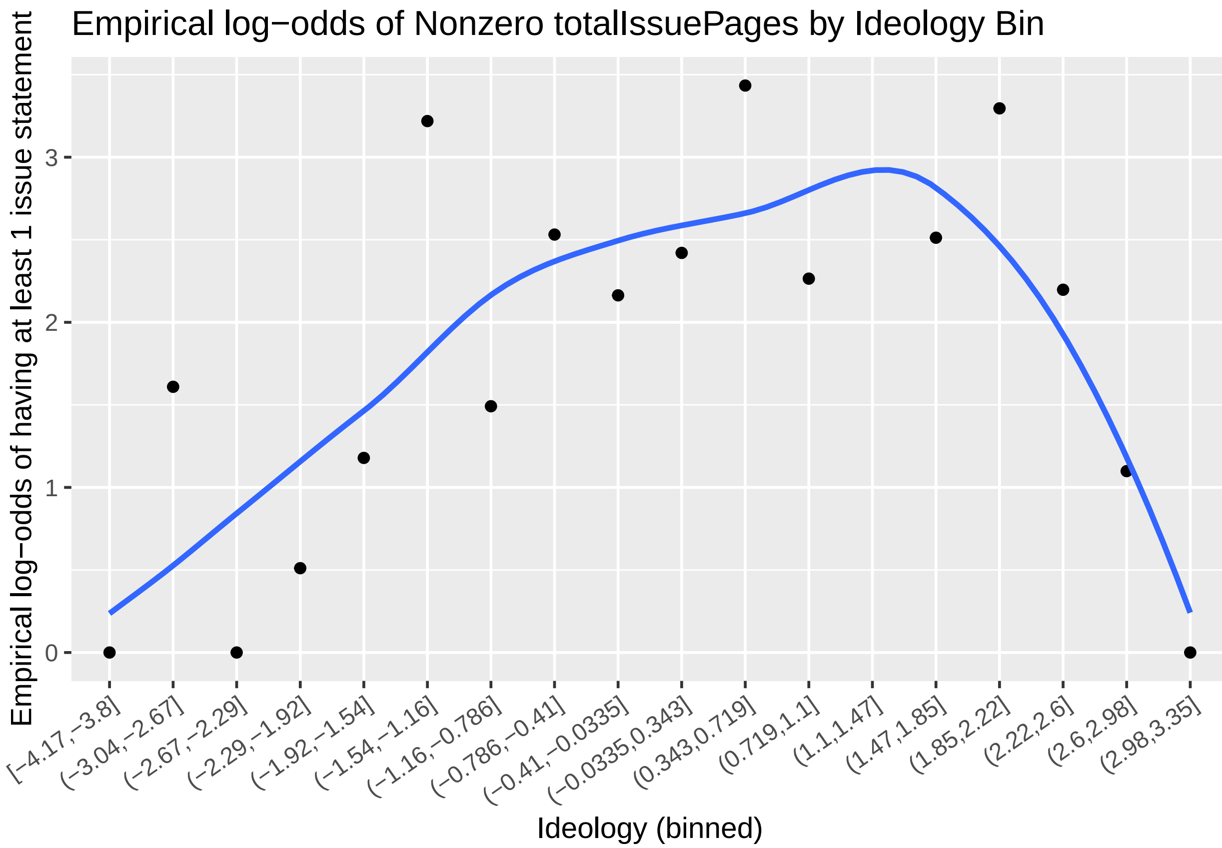


There is a substantial mass of candidates with totalIssuePages = 0, far more than a standard Poisson model would typically generate given the distribution of the positive counts. Among candidates who do comment on issues, the number of issue pages varies widely and is right-skewed, resembling a Poisson-like count distribution but clearly conditional on being nonzero. This suggests a two-stage process: First, whether a *candidate chooses to comment on any issue at all (zero vs nonzero). Second, how many issues they comment on given that they comment on at least one.*

A zero-inflated Poisson (ZIP) model is not appropriate here because a ZIP model assumes that zeros come from two latent sources: "Structural zeros" (cases that will always be zero), and "Sampling zeros" (cases that follow the Poisson process but happened to be zero this time). That assumption does not fit the political context of this data well. Here, a candidate with zero issue statements is simply silent, not a fundamentally different type of candidate who could never comment. There is no strong theoretical reason to believe there are two hidden kinds of zero-issue candidates. The hurdle model, by contrast, treats all zeros as the same meaningful outcome, which is much more consistent with the research question.

b. **Create a plot of the empirical log odds of having at least one issue statement by ideology. You may want to group ideology values first. What can you conclude from this plot?**

```r
amb2 <- amb |>
  filter(!is.na(ideology)) |>
  mutate(
    nonzero = as.integer(totalIssuePages > 0),
    ideologybin = ggplot2::cut_interval(ideology, n = 20)
  )

emp_logodds <- amb2 |>
  group_by(ideologybin) |>
  summarise(
    n = n(),
    prop_nonzero = mean(nonzero),
    .groups = "drop"
  ) |>
  # avoid infinite log-odds if prop is 0 or 1
  mutate(
    prop_nonzero_adj = pmin(pmax(prop_nonzero, 1/(2*n)), 1 - 1/(2*n)),
    log_odds = log(prop_nonzero_adj / (1 - prop_nonzero_adj))
  )

ggplot(emp_logodds, aes(x = ideologybin, y = log_odds, group = 1)) +
  geom_point() +
  geom_smooth(se=FALSE) +
  labs(
    x = "Ideology (binned)",
    y = "Empirical log-odds of having at least 1 issue statement",
    title = "Empirical log-odds of Nonzero totalIssuePages by Ideology Bin"
  ) +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



The plot shows that the empirical log-odds of having at least one issue statement vary non-linearly with ideology. Candidates with moderate ideology values—particularly those near the center to moderately right-leaning bins—have the highest log-odds of commenting on at least one issue, indicating they are most likely to cross the zero hurdle. In contrast, candidates at the ideological extremes, especially at the far right end of the scale, exhibit substantially lower log-odds, suggesting a greater likelihood of remaining silent. Overall, the relationship between ideology and the probability of issuing at least one statement is not monotonic, which supports the use of a flexible model such as a hurdle model rather than a simple Poisson regression.

c. **Create a hurdle model with ideology and democrat as predictors in both parts. Display the model. Interpret ideology in both parts of the model.**

```r
# Fit hurdle model (Poisson count part by default)
m1 <- hurdle(
  totalIssuePages ~ ideology + democrat | ideology + democrat,
  data = amb,
  dist = "poisson"
)

summary(m1)
```

```
##
## Call:
## hurdle(formula = totalIssuePages ~ ideology + democrat | ideology + democrat,
##     data = amb, dist = "poisson")
##
## Pearson residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -2.5897 -0.7846 -0.1034  0.7202 16.9177
##
## Count model coefficients (truncated poisson with log link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.093167   0.026736  78.289   <2e-16 ***
## ideology    -0.005902   0.020820  -0.283    0.777
## democrat     0.041379   0.040008   1.034    0.301
## Zero hurdle model coefficients (binomial with logit link):
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   2.1266     0.2247   9.465  < 2e-16 ***
## ideology      0.5746     0.1667   3.446 0.000568 ***
## democrat      0.4279     0.3494   1.225 0.220683
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Number of iterations in BFGS optimization: 8
## Log-likelihood: -2131 on 6 Df
```

Zero hurdle part (binary model: having at least one issue statement): In the zero hurdle component, ideology has a positive and statistically significant coefficient (estimate: 0.575, $p < 0.001$). This indicates that as a candidate's ideology score increases, the log-odds of having at least one issue statement increase, holding party affiliation constant. Substantively, candidates with higher ideology values are more likely to cross the hurdle from silence to engagement by commenting on at least one issue. This suggests that ideology plays an important role in explaining whether candidates choose to speak at all on policy issues.

Count part (positive counts only): In the count component, ideology has a coefficient very close to zero and is not statistically significant (estimate: -0.006, $p = 0.777$). This implies that among candidates who have already commented on at least one issue, ideology is not meaningfully associated with the number of issue pages they produce. In other words, once candidates decide to engage, ideology does not explain variation in how many issues they comment on.

d. **Repeat (d), but include an interaction in both parts. Interpret the interaction in the zero hurdle part of the model.**

```r
m2 <- hurdle(
  totalIssuePages ~ ideology * democrat | ideology * democrat,
  data = amb,
  dist = "poisson"
)

summary(m2)
```

```
##
## Call:
## hurdle(formula = totalIssuePages ~ ideology * democrat | ideology * democrat,
##     data = amb, dist = "poisson")
##
## Pearson residuals:
##      Min       1Q   Median       3Q      Max
## -2.90849 -0.73294 -0.07367  0.66102 16.65303
##
## Count model coefficients (truncated poisson with log link):
##                   Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)        2.05817    0.03315  62.087   <2e-16 ***
## ideology           0.03387    0.03006   1.127   0.2598
## democrat           0.05736    0.04135   1.387   0.1654
## ideology:democrat -0.07592    0.04141  -1.833   0.0668 .
## Zero hurdle model coefficients (binomial with logit link):
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)         1.8129     0.2319   7.819 5.34e-15 ***
## ideology            1.3667     0.2795   4.890 1.01e-06 ***
## democrat            0.3581     0.3175   1.128 0.259249
## ideology:democrat  -1.3995     0.3731  -3.751 0.000176 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Number of iterations in BFGS optimization: 9
## Log-likelihood: -2122 on 8 Df
```

In the zero hurdle component, the interaction between ideology and party affiliation is negative and statistically significant, indicating that the relationship between ideology and the probability of having at least one issue statement differs by party. Specifically, while higher ideology values are associated with a higher likelihood of commenting on at least one issue for Republicans (the reference group), this positive association is substantially weaker for Democrats. In fact, the negative interaction implies that as ideology increases, Democrats experience a much smaller increase—or even a decrease—in the odds of crossing the hurdle compared to Republicans. Thus, ideology is a strong predictor of whether candidates speak at all, but its effect depends importantly on party affiliation, with ideology playing a more prominent role in predicting non-silence among Republicans than among Democrats.

Grading

Total	**39**
Ex 1	4
Ex 2	6
Ex 3	8
Ex 4	8
Ex 5	10
Workflow & formatting	3

The "Workflow & formatting" grade is to based on the organization of the assignment write up along with the reproducible workflow. This includes having an organized write up with neat and readable headers, code, and narrative, including properly rendered mathematical notation. It also includes having a reproducible .Rmd document that can be rendered to reproduce the submitted PDF.

Extra resources and hints for Exercise 5 (Hurdle Problem)

1. There is an article on hurdle models that has helped some students that can be found here:

https://jsdajournal.springeropen.com/articles/10.1186/s40488-021-00121-4

2. There are two parts to the hurdle model, the count part and the binary part of the model and how to run it can be found in the R documentation.

3. You are not required to account for overdispersion (or check for it), however, if you do, this is great and just please make sure to think through how to do this properly as it involves integrating multiple parts of the Poisson lectures.

4. How do we handle the data in 5b? Do we bin it?

Yes, you should bin it or group it using the function cut_interval.

For example, something like this might help:

ideologybin = cut_interval(ideology, n=5)), where the number of bins of 5 was chosen empirically by playing around with the data.

References